# Projet_MA_vins_Gaudin_Le-Meur

2024

## Contents

**Introduction** **2**
Sujet . . . 2
Importation des données . . . 2

**I Analyse univariée** **3**
Summary . . . 3
Histogrammes . . . 4
Boxplots . . . 5
Interprétation . . . 7

**II Analyse multivariée** **7**
Coéfficients de corrélation . . . 7
ACP . . . 13

**III Classification des pays** **16**
Classification ascendante hierarchique . . . 16
K-means . . . 22

# Introduction

## Sujet

**Fichier** : *vins.xls*

La consommation de plusieurs types de vins (exprimee en KiloLitres/an) est observee dans plusieurs pays.

**Objectif** : Analyse descriptive multivariée, classification des pays.

## Importation des données

Dans un premier temps on importe les données en ajoutant les noms de vins comme titre des lignes:

```
vins <- read_excel("vins.xls")
vins=t(vins)
d<-vins[2:9,]
colnames(d)=vins[1,1:18]
d<-as.data.frame(d)
nomcol=colnames(d)
for (i in nomcol){
  d[,i]=as.integer(d[,i])
}
```

```
head(d)
```

```
##           CHMPAGNE MOUSSEAU_AOC MOUSSEAU_SIMPLE ALSACE GIRONDE BOJOLAIS
## BELGIQUE      7069         2436            3066   2422   22986    17465
## NEDERLAND     3786          586             290   1999   22183    19840
## RFA          12578         2006           10439  17183   21023    72977
## ITALIE        8037           30            1413     57      56     2364
## UK           13556         1217            7214   1127   30025    39919
## SUISSE        9664          471             112    600    6544    17327
##           BORDEAUX  RHON ANJOU AOC_AUTRES VDQS AUTRE_VDQS PROVENCE MUSCAT RHONE
## BELGIQUE      3784  7950  2587      17200 1976      38747     1375   2016   785
## NEDERLAND     2339 10537   600      22806 1029      19151     1150   2908  1648
## RFA           4828  7552  2101      15979 1346     191140     2514   1529  1009
## ITALIE          98    24     0         50    0       7992        0      0     6
## UK            7885  8172  7582      20004 2258     101108      284  12891   775
## SUISSE        3191 11691   143       1279  212       1029      401     18   643
##           AOC_FORT AUTRE_FORT TRES_FORT
## BELGIQUE       160         24      2415
## NEDERLAND      246       1533        74
## RFA            135        160       208
## ITALIE           8          0         8
## UK            1177        480      1705
## SUISSE          26          0        12
```

```
str(d)
```

```
## 'data.frame':    8 obs. of  18 variables:
##  $ CHMPAGNE       : int  7069 3786 12578 8037 13556 9664 10386 206
##  $ MOUSSEAU_AOC   : int  2436 586 2006 30 1217 471 997 51
##  $ MOUSSEAU_SIMPLE: int  3066 290 10439 1413 7214 112 3788 330
##  $ ALSACE         : int  2422 1999 17183 57 1127 600 408 241
##  $ GIRONDE        : int  22986 22183 21023 56 30025 6544 13114 3447
##  $ BOJOLAIS       : int  17465 19840 72977 2364 39919 17327 17487 2346
##  $ BORDEAUX       : int  3784 2339 4828 98 7885 3191 11791 1188
```

```
##  $ RHON           : int  7950 10537 7552 24 8172 11691 1369 1798
##  $ ANJOU          : int  2587 600 2101 0 7582 143 872 131
##  $ AOC_AUTRES     : int  17200 22806 15979 50 20004 1279 4016 944
##  $ VDQS           : int  1976 1029 1346 0 2258 212 1017 487
##  $ AUTRE_VDQS     : int  38747 19151 191140 7992 101108 1029 26192 38503
##  $ PROVENCE       : int  1375 1150 2514 0 284 401 9 236
##  $ MUSCAT         : int  2016 2908 1529 0 12891 18 716 653
##  $ RHONE          : int  785 1648 1009 6 775 643 542 35
##  $ AOC_FORT       : int  160 246 135 8 1177 26 7 0
##  $ AUTRE_FORT     : int  24 1533 160 0 480 0 0 0
##  $ TRES_FORT      : int  2415 74 208 8 1705 12 36 47
```

# I Analyse univariée

## Summary

Pour commencer l'analyse univarié on regarde le *summary*:

```
stat_uni = function(x)
{
return(c(

min(x, na.rm=TRUE),
max(x, na.rm=TRUE),
mean(x, na.rm=TRUE),
median(x, na.rm=TRUE),
quantile(x, 0.25, na.rm=TRUE),
quantile(x,0.75, na.rm=TRUE),
sd(x, na.rm=TRUE)))
}
res_stat_uni = apply(d, 2, "stat_uni")
row.names(res_stat_uni) = c( "Min", "Max", "Moyenne", "Mediane", "Quantile 1", "Quantile 3", "Ecart-typ
#print(round(t(res_stat_uni),3))
knitr::kable(round(t(res_stat_uni),3), format = "markdown", align = 'r')
```

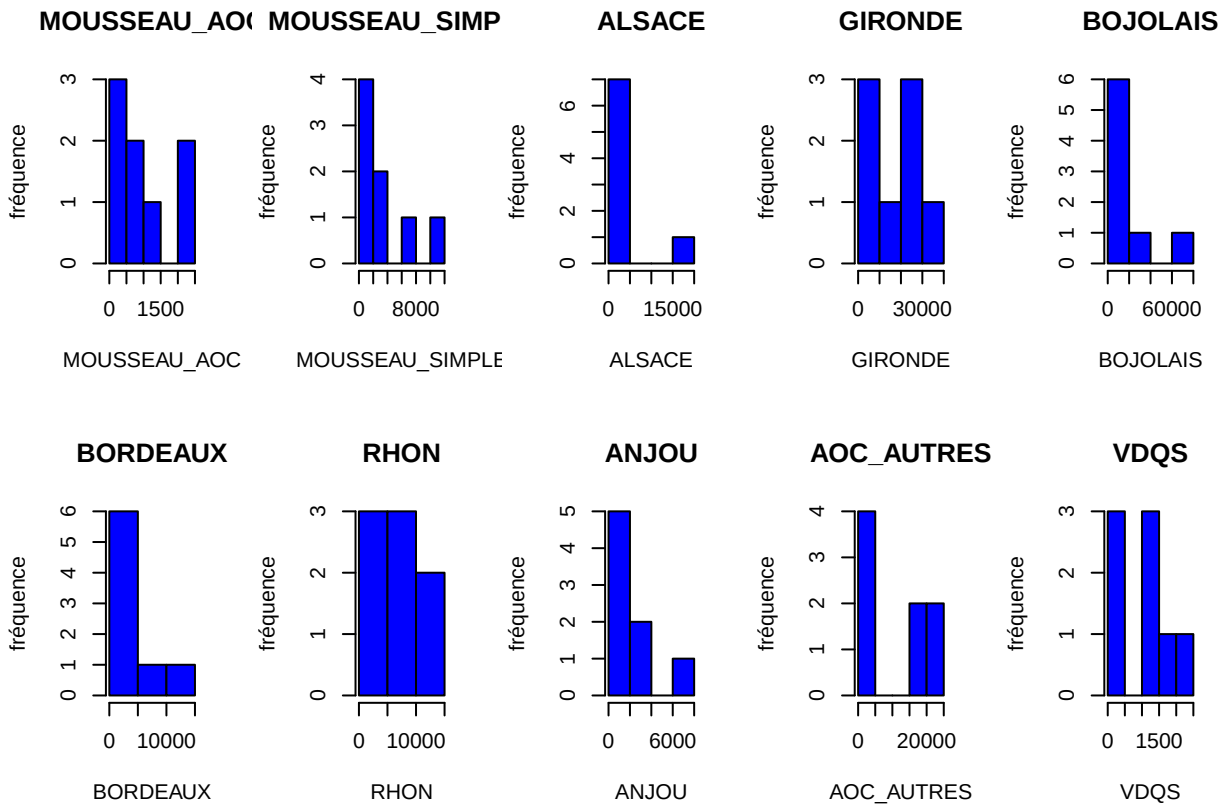
| | Min | Max | Moyenne | Mediane | Quantile 1 | Quantile 3 | Ecart-type |
|---|---|---|---|---|---|---|---|
| CHMPAGNE | 206 | 13556 | 8160.250 | 8850.5 | 6248.25 | 10934.00 | 4463.439 |
| MOUSSEAU_AOC | 30 | 2436 | 974.250 | 791.5 | 366.00 | 1414.25 | 879.014 |
| MOUSSEAU_SIMPLE | 112 | 10439 | 3331.500 | 2239.5 | 320.00 | 4644.50 | 3744.993 |
| ALSACE | 57 | 17183 | 3004.625 | 863.5 | 366.25 | 2104.75 | 5791.044 |
| GIRONDE | 56 | 30025 | 14922.250 | 17068.5 | 5769.75 | 22383.75 | 10751.786 |
| BOJOLAIS | 2346 | 72977 | 23715.625 | 17476.0 | 13586.25 | 24859.75 | 23100.669 |
| BORDEAUX | 98 | 11791 | 4388.000 | 3487.5 | 2051.25 | 5592.25 | 3814.767 |
| RHON | 24 | 11691 | 6136.625 | 7751.0 | 1690.75 | 8763.25 | 4449.187 |
| ANJOU | 0 | 7582 | 1752.000 | 736.0 | 140.00 | 2222.50 | 2540.963 |
| AOC_AUTRES | 50 | 22806 | 10284.750 | 9997.5 | 1195.25 | 17901.00 | 9590.853 |
| VDQS | 0 | 2258 | 1040.625 | 1023.0 | 418.25 | 1503.50 | 804.176 |
| AUTRE_VDQS | 1029 | 191140 | 52982.750 | 32347.5 | 16361.25 | 54337.25 | 63697.361 |
| PROVENCE | 0 | 2514 | 746.125 | 342.5 | 179.25 | 1206.25 | 876.432 |
| MUSCAT | 0 | 12891 | 2591.375 | 1122.5 | 494.25 | 2239.00 | 4280.421 |
| RHONE | 6 | 1648 | 680.375 | 709.0 | 415.25 | 841.00 | 528.614 |
| AOC_FORT | 0 | 1177 | 219.875 | 80.5 | 7.75 | 181.50 | 397.078 |
| AUTRE_FORT | 0 | 1533 | 274.625 | 12.0 | 0.00 | 240.00 | 534.965 |

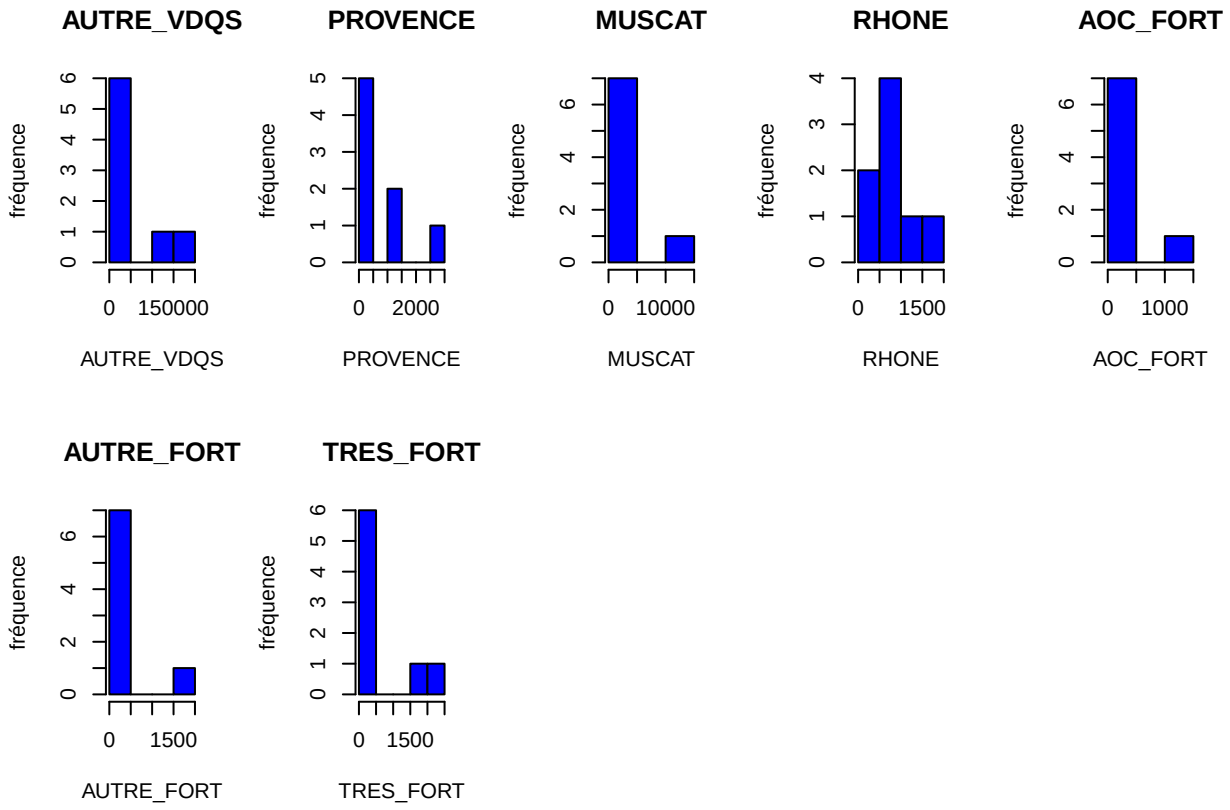
| | Min | Max | Moyenne | Mediane | Quantile 1 | Quantile 3 | Ecart-type |
|---|---|---|---|---|---|---|---|
| TRES_FORT | 8 | 2415 | 563.125 | 60.5 | 30.00 | 582.25 | 945.276 |

## Histogrammes

Esnuite on peut aussi observer les différents *Histogrammes*:

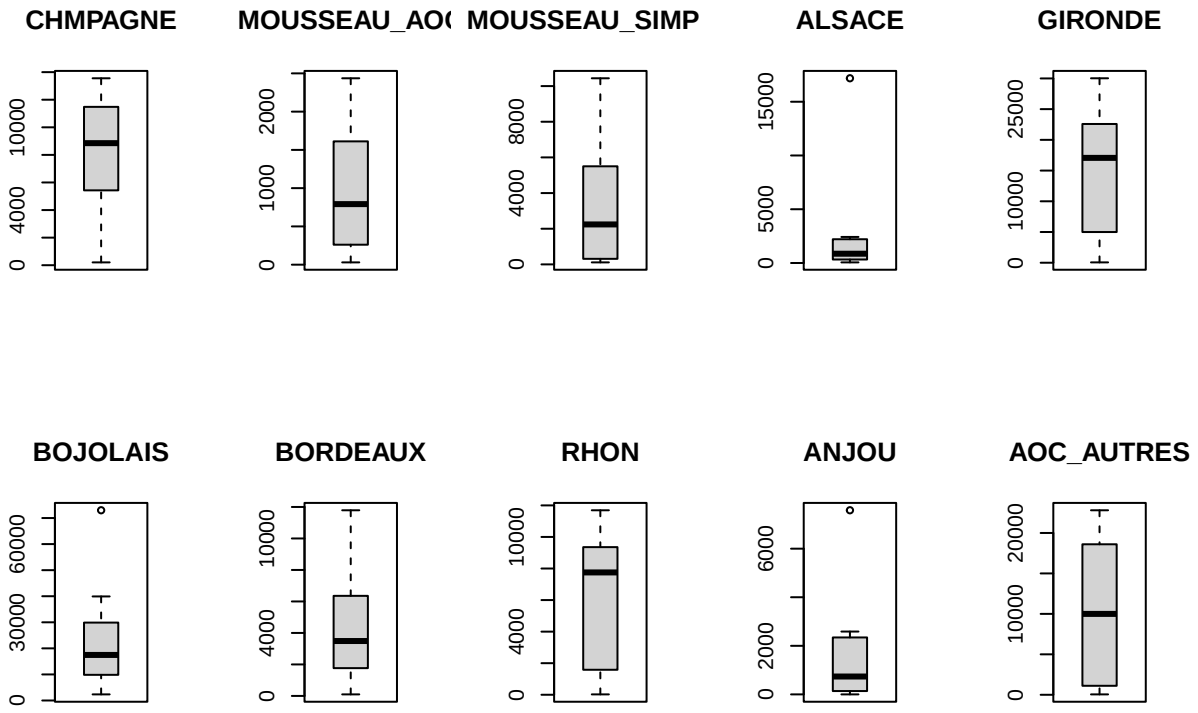
```
par(mfrow=c(2,5))
for(i in 2:ncol(d))
{
hist(d[,names(d)[i]], xlab = names(d)[i], ylab = "fréquence", col = "blue", main = names(d)[i])
}
```

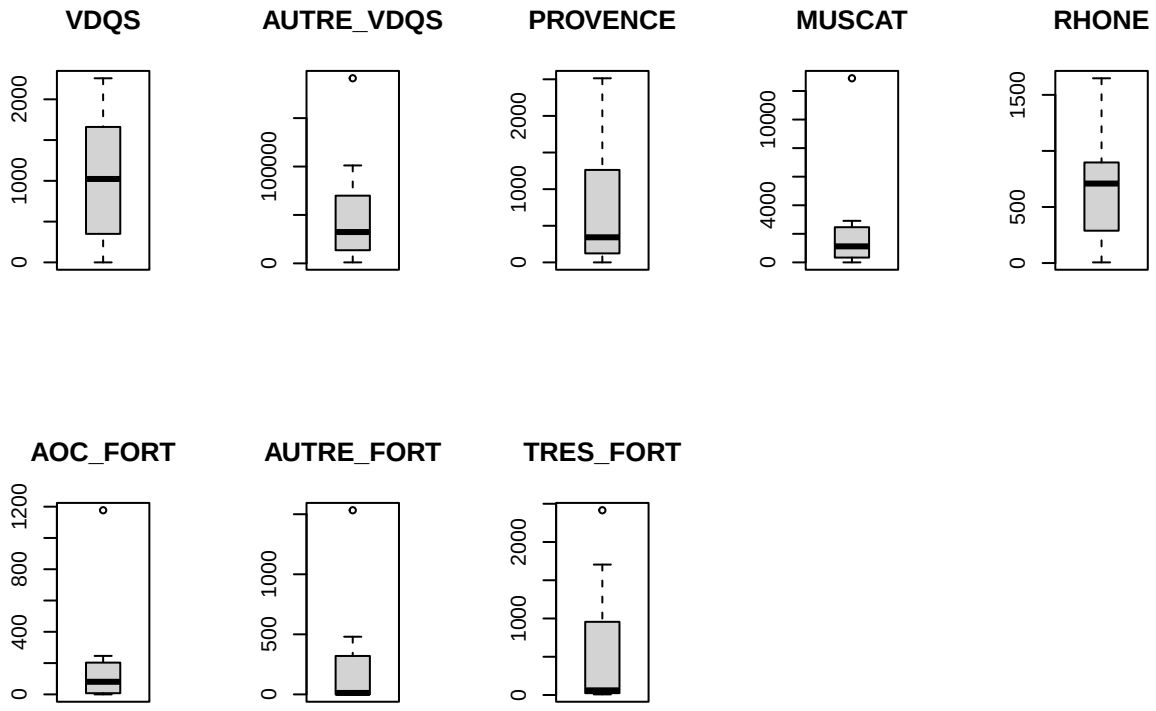





## Boxplots

Et enfin des *Boxplots* peuvent être crées :


CHMPAGNE
MOUSSEAU_AO(
MOUSSEAU_SIMP
ALSACE
GIRONDE
BOJOLAIS
BORDEAUX
RHON
ANJOU
AOC_AUTRES




### Interprétation

Dans un premier temps on observe qu'il n'y à pas de données manquantes. De plus les valeurs vont de *0* comme *minimum* à *191140* comme *maximum* (AUTRE_VDQS). Ensuite on remarque que La moyenne des mesures varie fortement selon les catégories de vin, allant de 0 à 52982.75, et de même la médiane varie aussi en fonction des catégories de vin, ce qui indique une certaine asymétrie dans les distributions. Pour terminer l'écart-type représente la dispersion des données autour de la moyenne, ces valeurs varient considérablement selon les catégories de vin, suggérant des degrés différents de dispersion ou de variabilité des données.

**Conclusion**: Pour résumer, l'analyse univariée met en évidence des écarts significatifs dans les mesures statistiques observées pour chaque catégorie de vin. Ces différences peuvent être attribuées aux caractéristiques uniques et aux volumes de production spécifiques à chaque type de vin.

## II Analyse multivariée

### Coéfficients de corrélation

```
cor=cov.wt(d, method="ML", cor=TRUE)$cor
cor<-as.data.frame(cor)
kable(round(cor[1:9],2), format="markdown")
```

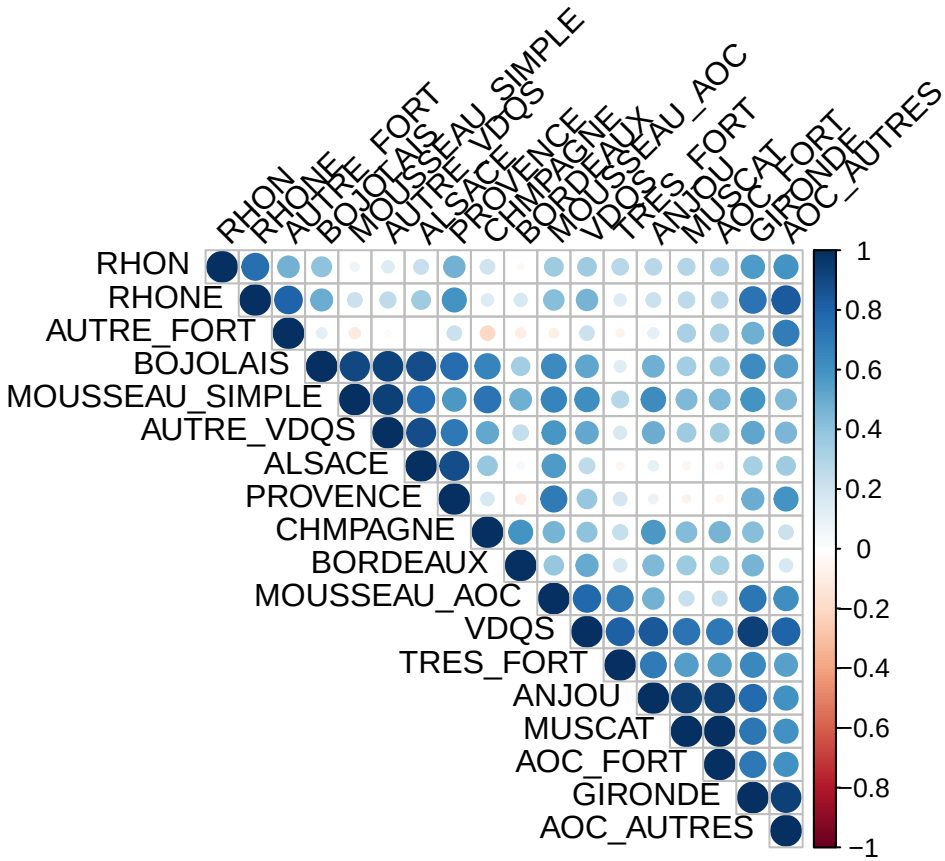
| | CHMPAGNE | MOUSSEAU_AOC | MOUSSEAU_SIMPLE | ALSACE | GIRONDE | BOJOLAIS | BORDEAUX | RHON | ANJOU |
|---|---|---|---|---|---|---|---|---|---|
| CHMPAGNE | 1.00 | 0.46 | 0.74 | 0.39 | 0.43 | 0.66 | 0.60 | 0.21 | 0.58 |
| MOUSSEAU_AOC | 0.46 | 1.00 | 0.66 | 0.57 | 0.72 | 0.63 | 0.38 | 0.36 | 0.47 |
| MOUSSEAU_SIMPLE | 0.74 | 0.66 | 1.00 | 0.77 | 0.59 | 0.91 | 0.48 | 0.08 | 0.62 |

| | CHMPAGNE | MOUSSEAU_AOC | MOUSSEAU_SIMPLE | ALSACE | GIRONDE | BOJOLAIS | BORDEAUX | RHON | ANJOU |
|---|---|---|---|---|---|---|---|---|---|
| ALSACE | 0.39 | 0.57 | 0.77 | 1.00 | 0.34 | 0.89 | 0.05 | 0.22 | 0.10 |
| GIRONDE | 0.43 | 0.72 | 0.59 | 0.34 | 1.00 | 0.63 | 0.46 | 0.56 | 0.78 |
| BOJOLAIS | 0.66 | 0.63 | 0.91 | 0.89 | 0.63 | 1.00 | 0.34 | 0.41 | 0.48 |
| BORDEAUX | 0.60 | 0.38 | 0.48 | 0.05 | 0.46 | 0.34 | 1.00 | -0.04 | 0.45 |
| RHON | 0.21 | 0.36 | 0.08 | 0.22 | 0.56 | 0.41 | -0.04 | 1.00 | 0.28 |
| ANJOU | 0.58 | 0.47 | 0.62 | 0.10 | 0.78 | 0.48 | 0.45 | 0.28 | 1.00 |
| AOC_AUTRES | 0.21 | 0.61 | 0.45 | 0.36 | 0.93 | 0.56 | 0.18 | 0.60 | 0.60 |
| VDQS | 0.41 | 0.79 | 0.62 | 0.25 | 0.93 | 0.53 | 0.51 | 0.35 | 0.84 |
| AUTRE_VDQS | 0.51 | 0.58 | 0.93 | 0.88 | 0.53 | 0.93 | 0.24 | 0.15 | 0.50 |
| PROVENCE | 0.18 | 0.71 | 0.57 | 0.89 | 0.50 | 0.76 | -0.10 | 0.47 | 0.08 |
| MUSCAT | 0.43 | 0.23 | 0.44 | -0.06 | 0.72 | 0.35 | 0.37 | 0.29 | 0.95 |
| RHONE | 0.15 | 0.43 | 0.22 | 0.36 | 0.74 | 0.50 | 0.17 | 0.76 | 0.22 |
| AOC_FORT | 0.46 | 0.22 | 0.45 | -0.05 | 0.71 | 0.36 | 0.34 | 0.33 | 0.95 |
| AUTRE_FORT | -0.21 | -0.08 | -0.12 | -0.01 | 0.49 | 0.11 | -0.10 | 0.48 | 0.12 |
| TRES_FORT | 0.23 | 0.70 | 0.28 | -0.05 | 0.65 | 0.14 | 0.18 | 0.28 | 0.70 |

```
kable(round(cor[10:18],2), format="markdown")
```

| | AOC_AUTRES | VDQS | AUTRE_VDQS | PROVENCE | MUSCAT | RHONE | AOC_FORT | AUTRE_FORT | TRES_FORT |
|---|---|---|---|---|---|---|---|---|---|
| CHMPAGNE | 0.21 | 0.41 | 0.51 | 0.18 | 0.43 | 0.15 | 0.46 | -0.21 | 0.23 |
| MOUSSEAU_AOC | 0.61 | 0.79 | 0.58 | 0.71 | 0.23 | 0.43 | 0.22 | -0.08 | 0.70 |
| MOUSSEAU_SIMPLE | 0.45 | 0.62 | 0.93 | 0.57 | 0.44 | 0.22 | 0.45 | -0.12 | 0.28 |
| ALSACE | 0.36 | 0.25 | 0.88 | 0.89 | -0.06 | 0.36 | -0.05 | -0.01 | -0.05 |
| GIRONDE | 0.93 | 0.93 | 0.53 | 0.50 | 0.72 | 0.74 | 0.71 | 0.49 | 0.65 |
| BOJOLAIS | 0.56 | 0.53 | 0.93 | 0.76 | 0.35 | 0.50 | 0.36 | 0.11 | 0.14 |
| BORDEAUX | 0.18 | 0.51 | 0.24 | -0.10 | 0.37 | 0.17 | 0.34 | -0.10 | 0.18 |
| RHON | 0.60 | 0.35 | 0.15 | 0.47 | 0.29 | 0.76 | 0.33 | 0.48 | 0.28 |
| ANJOU | 0.60 | 0.84 | 0.50 | 0.08 | 0.95 | 0.22 | 0.95 | 0.12 | 0.70 |
| AOC_AUTRES | 1.00 | 0.80 | 0.46 | 0.60 | 0.60 | 0.84 | 0.60 | 0.69 | 0.54 |
| VDQS | 0.80 | 1.00 | 0.52 | 0.39 | 0.73 | 0.47 | 0.71 | 0.21 | 0.82 |
| AUTRE_VDQS | 0.46 | 0.52 | 1.00 | 0.71 | 0.35 | 0.26 | 0.35 | -0.03 | 0.17 |
| PROVENCE | 0.60 | 0.39 | 0.71 | 1.00 | -0.06 | 0.60 | -0.05 | 0.21 | 0.18 |
| MUSCAT | 0.60 | 0.73 | 0.35 | -0.06 | 1.00 | 0.27 | 1.00 | 0.33 | 0.56 |
| RHONE | 0.84 | 0.47 | 0.26 | 0.60 | 0.27 | 1.00 | 0.27 | 0.80 | 0.15 |
| AOC_FORT | 0.60 | 0.71 | 0.35 | -0.05 | 1.00 | 0.27 | 1.00 | 0.33 | 0.55 |
| AUTRE_FORT | 0.69 | 0.21 | -0.03 | 0.21 | 0.33 | 0.80 | 0.33 | 1.00 | -0.06 |
| TRES_FORT | 0.54 | 0.82 | 0.17 | 0.18 | 0.56 | 0.15 | 0.55 | -0.06 | 1.00 |

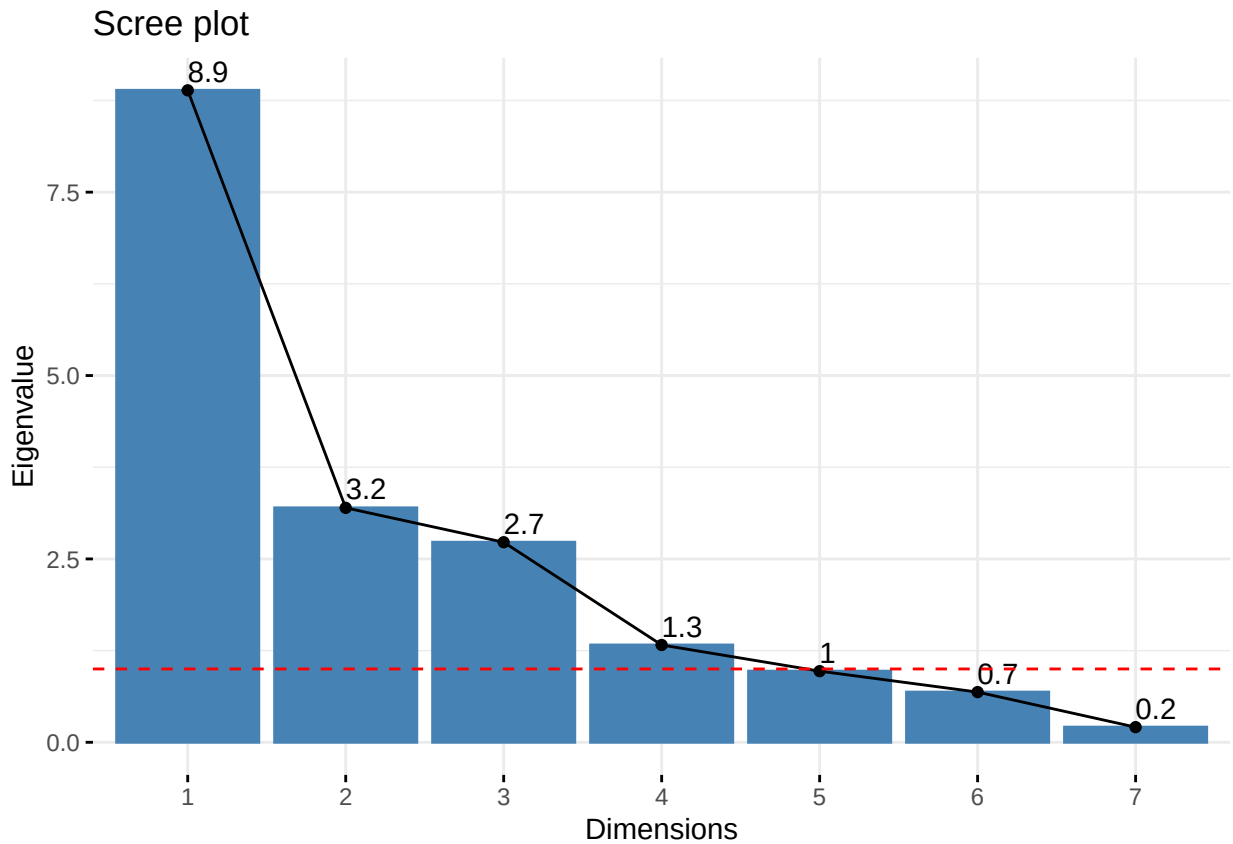
```
corr<-cor(d, method="pearson")
corrplot(corr, type="upper", order="hclust", tl.col="black", tl.srt=45)
```



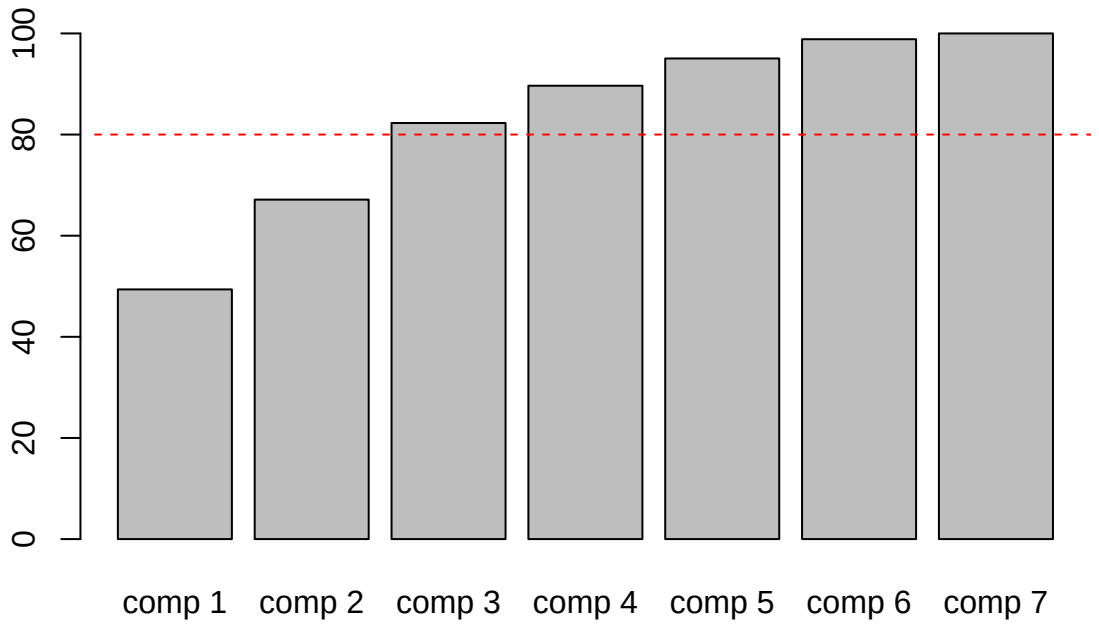

```
A = PCA(d, scale.unit=TRUE,graph=FALSE)
eig=A$eig

kable(eig, format="markdown")
```

| | eigenvalue | percentage of variance | cumulative percentage of variance |
|---|---|---|---|
| comp 1 | 8.8880543 | 49.378080 | 49.37808 |
| comp 2 | 3.1963075 | 17.757264 | 67.13534 |
| comp 3 | 2.7276039 | 15.153355 | 82.28870 |
| comp 4 | 1.3266750 | 7.370416 | 89.65911 |
| comp 5 | 0.9700696 | 5.389276 | 95.04839 |
| comp 6 | 0.6845571 | 3.803095 | 98.85149 |
| comp 7 | 0.2067326 | 1.148515 | 100.00000 |

```
library(factoextra)
fviz_eig(A,choice = "eigenvalue", addlabels = TRUE )+
geom_hline (yintercept = 1, linetype = 2, color = "red")
```




```
barplot(A$eig[,3]) +
lines(c(0,20),c(80,80),type = "l", lty=2,col="red")
```




```
## numeric(0)
```

On doit prendre 3 axes. Selon le critère de Kaiser, les 80% d'inertie totale sont atteints à partir de la 3ème dimension. On ne retient pas le critère du coude, car cela nous ferait prendre 2 dimensions, qui ne seraient pas suffisantes pour retranscrire l'information

```
kable(A$var$coord[,1:3], format="markdown", caption = "coord")
```

**Afficher les cordonnées, les corrélations, les cosinus carrés (qualité de la représentation) et les contributions de chaque variable (média publicitaire) par rapport aux cinq premières composantes principales.**

Table 5: coord

| | Dim.1 | Dim.2 | Dim.3 |
|---|---|---|---|
| CHMPAGNE | 0.6072688 | 0.1024555 | -0.5041422 |
| MOUSSEAU_AOC | 0.7775888 | 0.2470758 | -0.0814971 |
| MOUSSEAU_SIMPLE | 0.8021424 | 0.3733386 | -0.4236510 |
| ALSACE | 0.5591280 | 0.8134794 | 0.0078692 |
| GIRONDE | 0.9448574 | -0.2172973 | 0.1820483 |
| BOJOLAIS | 0.8234525 | 0.4984034 | -0.0819653 |
| BORDEAUX | 0.4479770 | -0.1548469 | -0.4511489 |
| RHON | 0.5193233 | -0.0518218 | 0.5809746 |
| ANJOU | 0.8057605 | -0.4599190 | -0.3146331 |
| AOC_AUTRES | 0.8500064 | -0.1539469 | 0.4599846 |

| | Dim.1 | Dim.2 | Dim.3 |
|---|---|---|---|
| VDQS | 0.8948636 | -0.2799860 | -0.0775227 |
| AUTRE_VDQS | 0.7475147 | 0.5065459 | -0.2339489 |
| PROVENCE | 0.6127968 | 0.6676495 | 0.3491175 |
| MUSCAT | 0.6967702 | -0.6123740 | -0.1569654 |
| RHONE | 0.6271885 | 0.0377170 | 0.7099313 |
| AOC_FORT | 0.6995089 | -0.5985468 | -0.1479583 |
| AUTRE_FORT | 0.3001127 | -0.2669118 | 0.8028566 |
| TRES_FORT | 0.5935437 | -0.4236271 | -0.1342321 |

```
#kable(A$var$cor[,1:3], format="markdown", caption = "cor")
kable(A$var$cos2[,1:3], format="markdown", caption = "cos2")
```

Table 6: cos2

| | Dim.1 | Dim.2 | Dim.3 |
|---|---|---|---|
| CHMPAGNE | 0.3687753 | 0.0104971 | 0.2541593 |
| MOUSSEAU_AOC | 0.6046444 | 0.0610465 | 0.0066418 |
| MOUSSEAU_SIMPLE | 0.6434325 | 0.1393817 | 0.1794802 |
| ALSACE | 0.3126241 | 0.6617487 | 0.0000619 |
| GIRONDE | 0.8927555 | 0.0472181 | 0.0331416 |
| BOJOLAIS | 0.6780740 | 0.2484060 | 0.0067183 |
| BORDEAUX | 0.2006834 | 0.0239776 | 0.2035353 |
| RHON | 0.2696967 | 0.0026855 | 0.3375315 |
| ANJOU | 0.6492500 | 0.2115255 | 0.0989940 |
| AOC_AUTRES | 0.7225110 | 0.0236996 | 0.2115859 |
| VDQS | 0.8007809 | 0.0783922 | 0.0060098 |
| AUTRE_VDQS | 0.5587782 | 0.2565887 | 0.0547321 |
| PROVENCE | 0.3755199 | 0.4457558 | 0.1218830 |
| MUSCAT | 0.4854887 | 0.3750019 | 0.0246381 |
| RHONE | 0.3933654 | 0.0014226 | 0.5040025 |
| AOC_FORT | 0.4893127 | 0.3582583 | 0.0218917 |
| AUTRE_FORT | 0.0900676 | 0.0712419 | 0.6445787 |
| TRES_FORT | 0.3522942 | 0.1794599 | 0.0180183 |

```
kable(A$var$contrib[,1:3], format="markdown", caption = "contrib")
```

Table 7: contrib

| | Dim.1 | Dim.2 | Dim.3 |
|---|---|---|---|
| CHMPAGNE | 4.149112 | 0.3284142 | 9.3180435 |
| MOUSSEAU_AOC | 6.802888 | 1.9099057 | 0.2435024 |
| MOUSSEAU_SIMPLE | 7.239295 | 4.3607099 | 6.5801421 |
| ALSACE | 3.517351 | 20.7035354 | 0.0022703 |
| GIRONDE | 10.044443 | 1.4772708 | 1.2150433 |
| BOJOLAIS | 7.629049 | 7.7716548 | 0.2463081 |
| BORDEAUX | 2.257900 | 0.7501644 | 7.4620561 |
| RHON | 3.034373 | 0.0840188 | 12.3746523 |
| ANJOU | 7.304749 | 6.6178071 | 3.6293383 |
| AOC_AUTRES | 8.129012 | 0.7414696 | 7.7572063 |

| | Dim.1 | Dim.2 | Dim.3 |
|---|---|---|---|
| VDQS | 9.009631 | 2.4525853 | 0.2203312 |
| AUTRE_VDQS | 6.286845 | 8.0276606 | 2.0066004 |
| PROVENCE | 4.224996 | 13.9459616 | 4.4685023 |
| MUSCAT | 5.462260 | 11.7323468 | 0.9032883 |
| RHONE | 4.425776 | 0.0445068 | 18.4778464 |
| AOC_FORT | 5.505284 | 11.2085054 | 0.8025964 |
| AUTRE_FORT | 1.013356 | 2.2288810 | 23.6316826 |
| TRES_FORT | 3.963681 | 5.6146018 | 0.6605896 |

## ACP

Axe 1 : forte contribution : Gironde, VDQS, AOC_Autres fort cos2->mieux représenté : Gironde, VDQS, AOC_Autres toute les coordonnées sont positives ->effet taille car toutes les coords sont positives

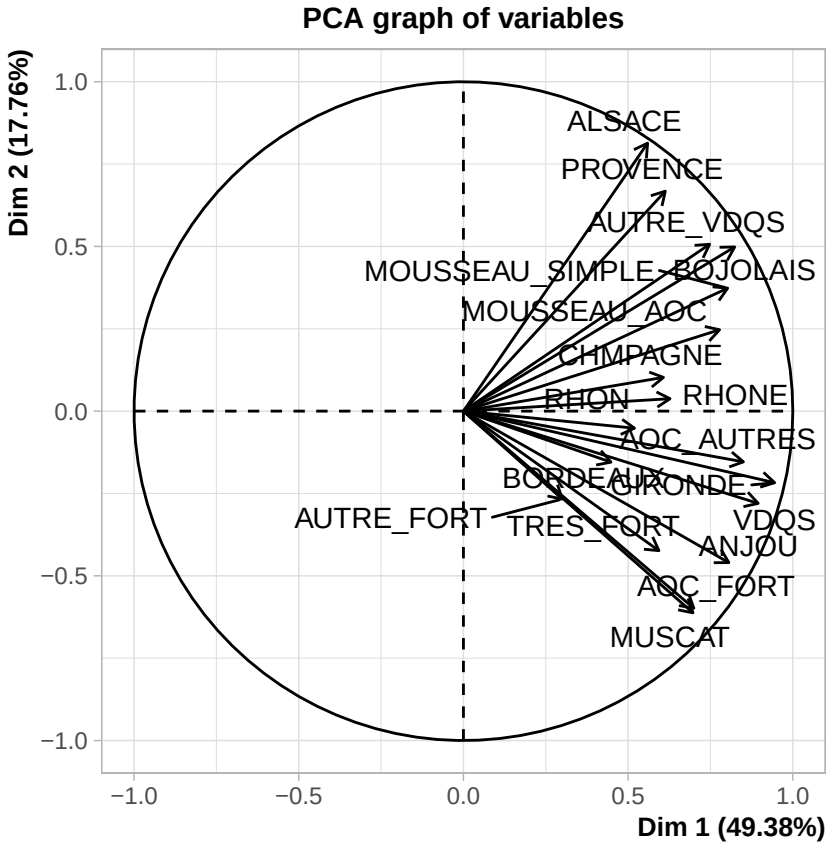
Axe2 : forte contribution: Alsace, Muscat, AOC_Fort, Provence fort cos2 : Alsace et Provence coord: Alsace et Provence positifs, opposé au muscat

Axe3: forte contrib : Rhon, Rhone, Autre_Fort fort cos2:Autre_Fort et Rhone coord : les 3 sont positives

**g. Représentation de la répartition des dépenses publicitaires (cercle des corrélations):**

**i. par rapport aux composantes principales 1 et 2 et analyser le graphique**

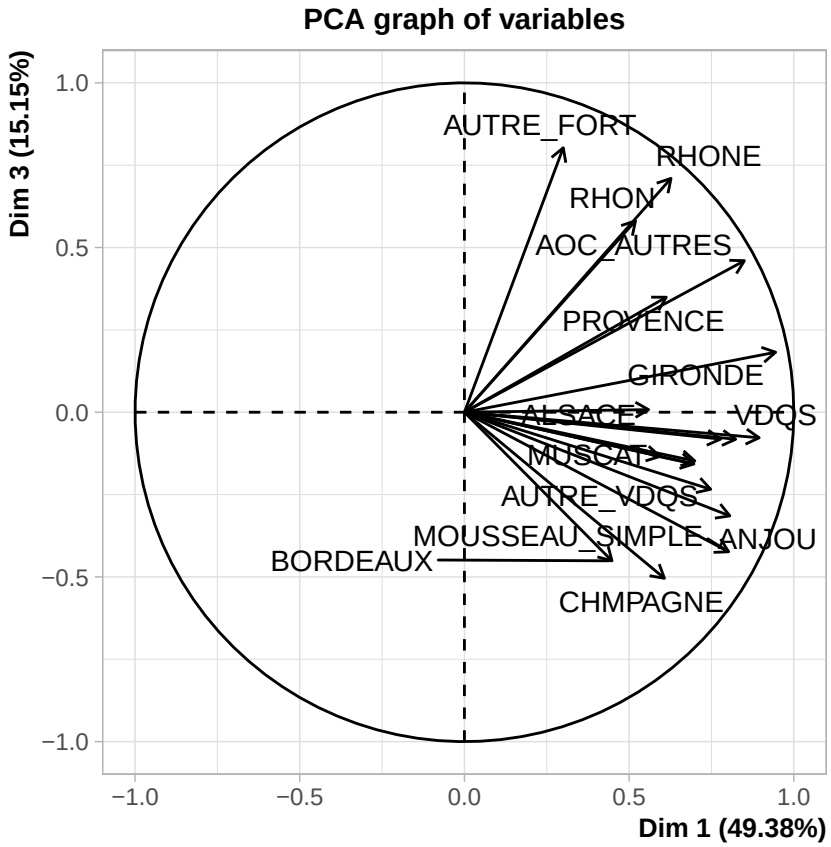
```
plot.PCA(A, axes=c(1,2), choix="var")
```



**ii. par rapport aux composantes principales 1 et 3 et analyser le graphique**

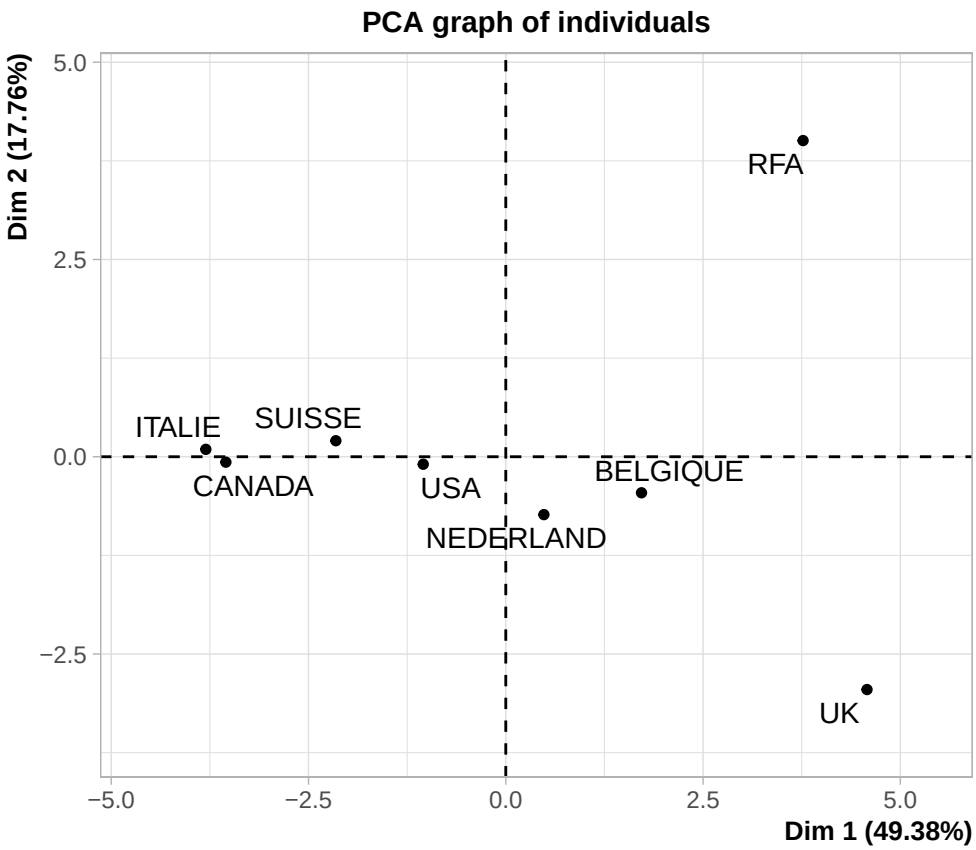
```
plot.PCA(A, axes=c(1,3), choix="var")
```

```
## Warning: ggrepel: 4 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```




**iii. Représentation de la répartition des pays par rapport aux composantes principales 1 et 2 et analyser le graphique.**

```
plot.PCA(A, axes=c(1,2), choix="ind")
```

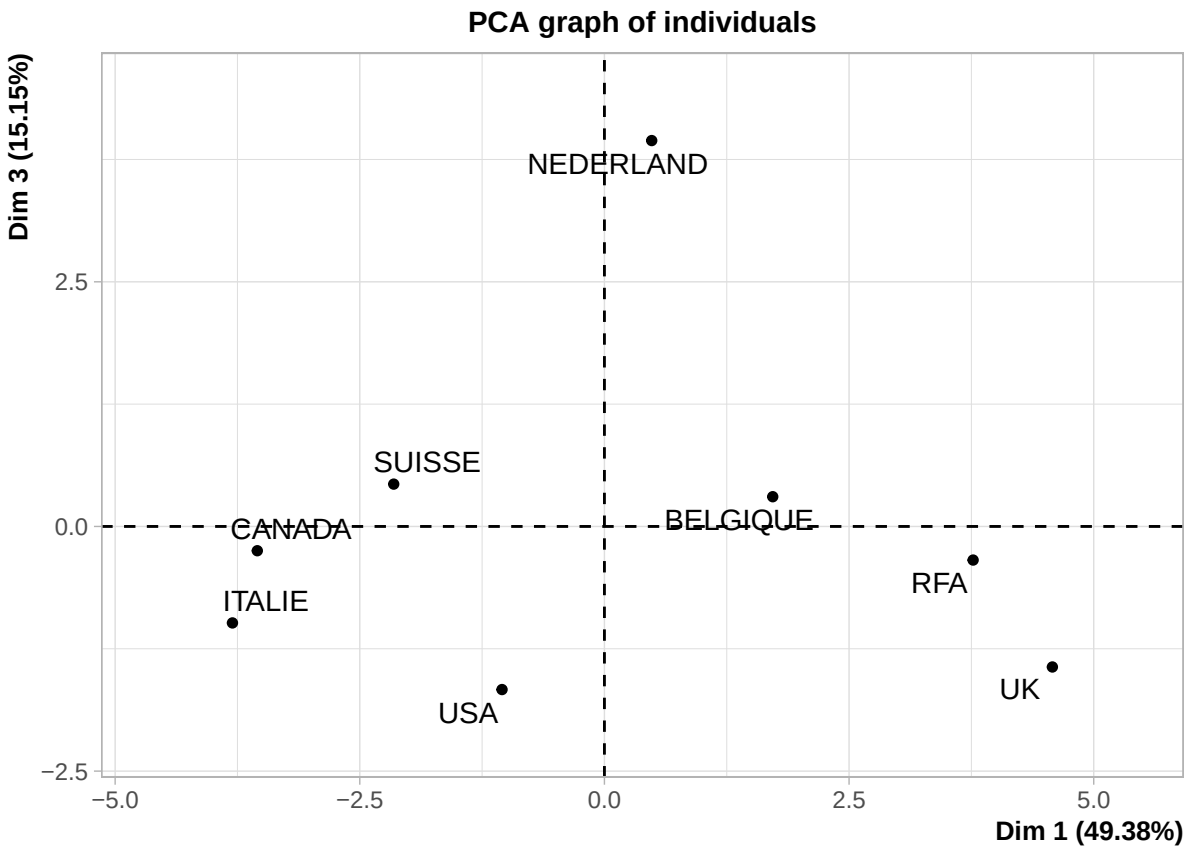



**Interprétation:**

Axe 1 :Oppose UK et RFA à Italie, Canada et Suisse Oppose les pays qui consomment beaucoup de vins à ceux qui en consomment peu

Axe2 : Oppose UK et RFA RFA consomme beaucoup de vins d'Alsace et de Provence, tandis que UK consomme plutot du muscat

```
plot.PCA(A, axes=c(1,3), choix="ind")
```




### Interprétation:

Axe 1 :Oppose UK et RFA à Italie, Canada et Suisse Oppose les pays qui consomment beaucoup de vins à ceux qui en consomment peu

Axe3 : Oppose Nederland à Italie, USA et UK

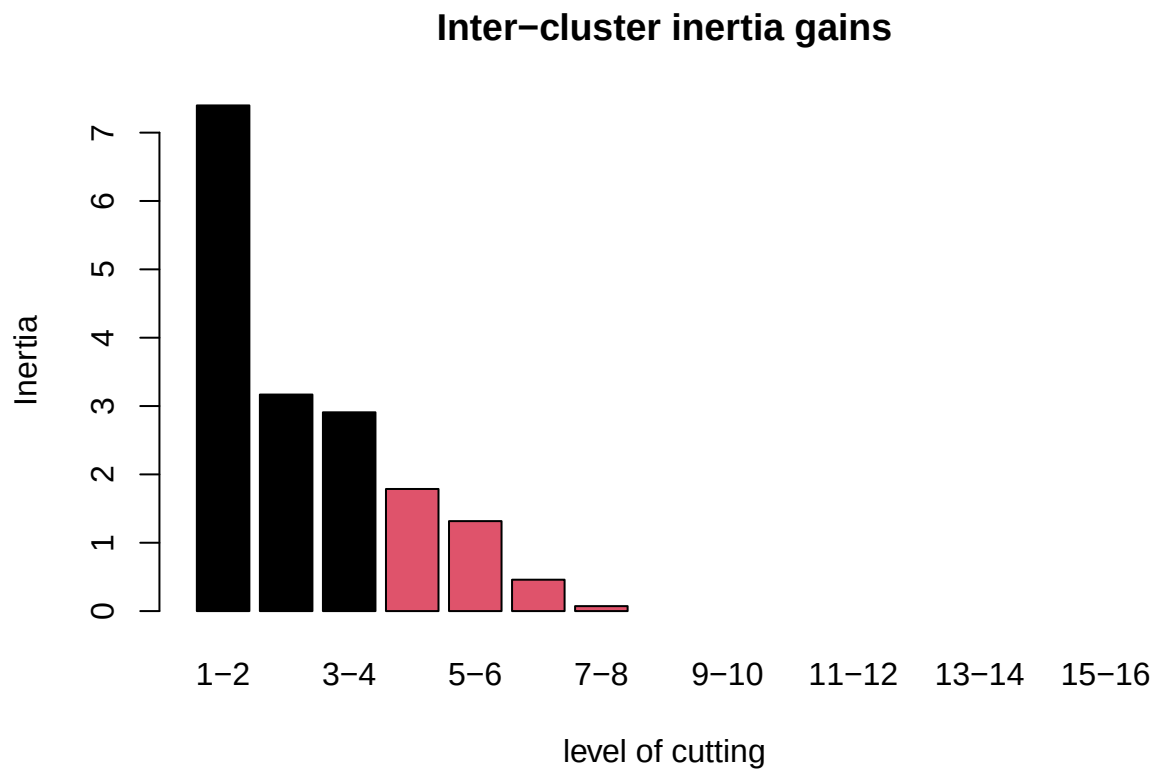
# III Classification des pays

## Classification ascendante hierarchique

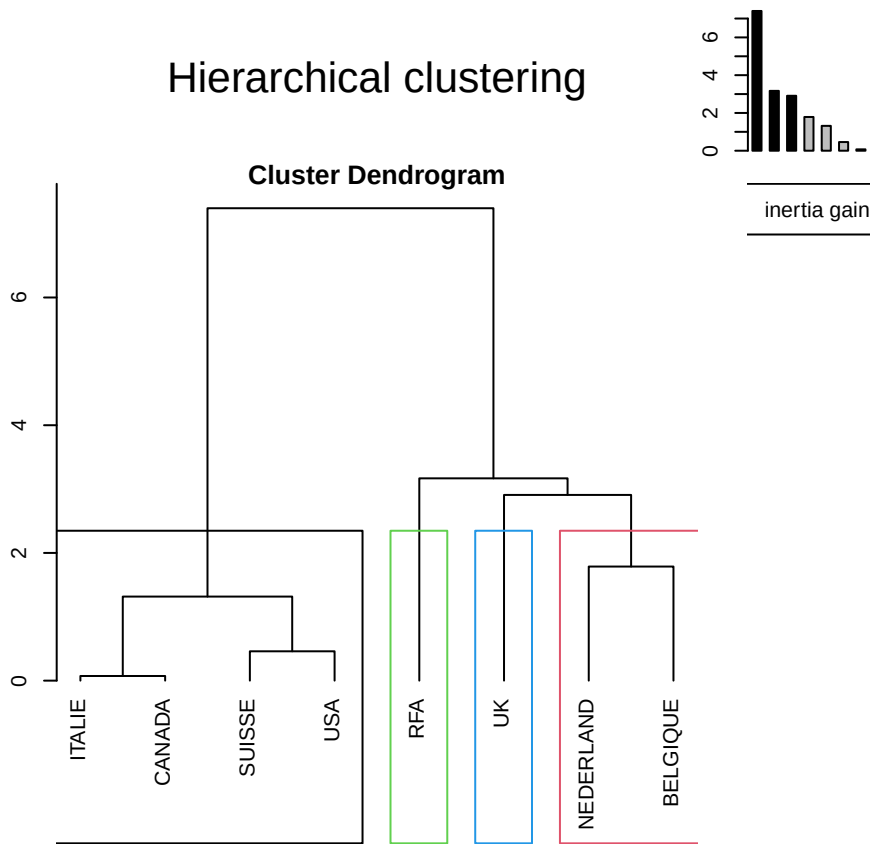
```
CAH=HCPC(A, nb.clust=-1, graph=FALSE)
CAH$data.clust$clust
```

```
## [1] 2 2 3 1 4 1 1 1
## Levels: 1 2 3 4
```

```
plot(CAH, choice="bar")
```




```
plot(CAH, choice="tree")
```



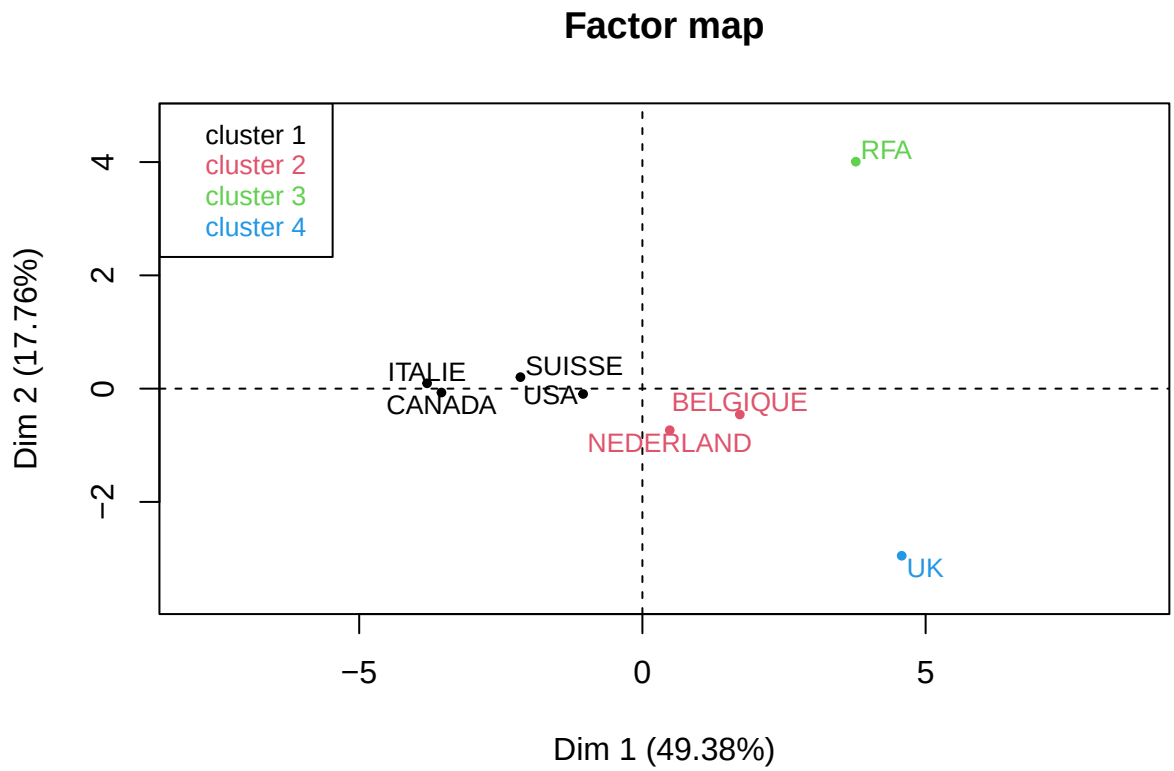

**Caractériser chaque classe à l'aide des composantes principales utilisées**

```
CAH$desc.axes
```

```
##
## Link between the cluster variable and the quantitative variables
## ================================================================
##             Eta2      P-value
## Dim.2 0.9961834 2.727729e-05
## Dim.1 0.9196585 1.177350e-02
##
## Description of each cluster by quantitative variables
## =====================================================
## $`1`
##          v.test Mean in category Overall mean sd in category Overall sd
## Dim.1 -2.340252        -2.637042 3.538836e-16        1.11217   2.981284
##          p.value
## Dim.1 0.01927072
##
## $`2`
##         v.test Mean in category Overall mean sd in category Overall sd
## Dim.3 1.964184         2.123658 6.106227e-16       1.819675   1.651546
##          p.value
## Dim.3 0.04950871
##
## $`3`
```

```
##         v.test Mean in category  Overall mean sd in category Overall sd p.value
## Dim.2 2.241248         4.006953 -1.960238e-16              0   1.787822 0.02501
##
## $`4`
## NULL
```

```
plot(CAH, choice="map", draw.tree = F)
```

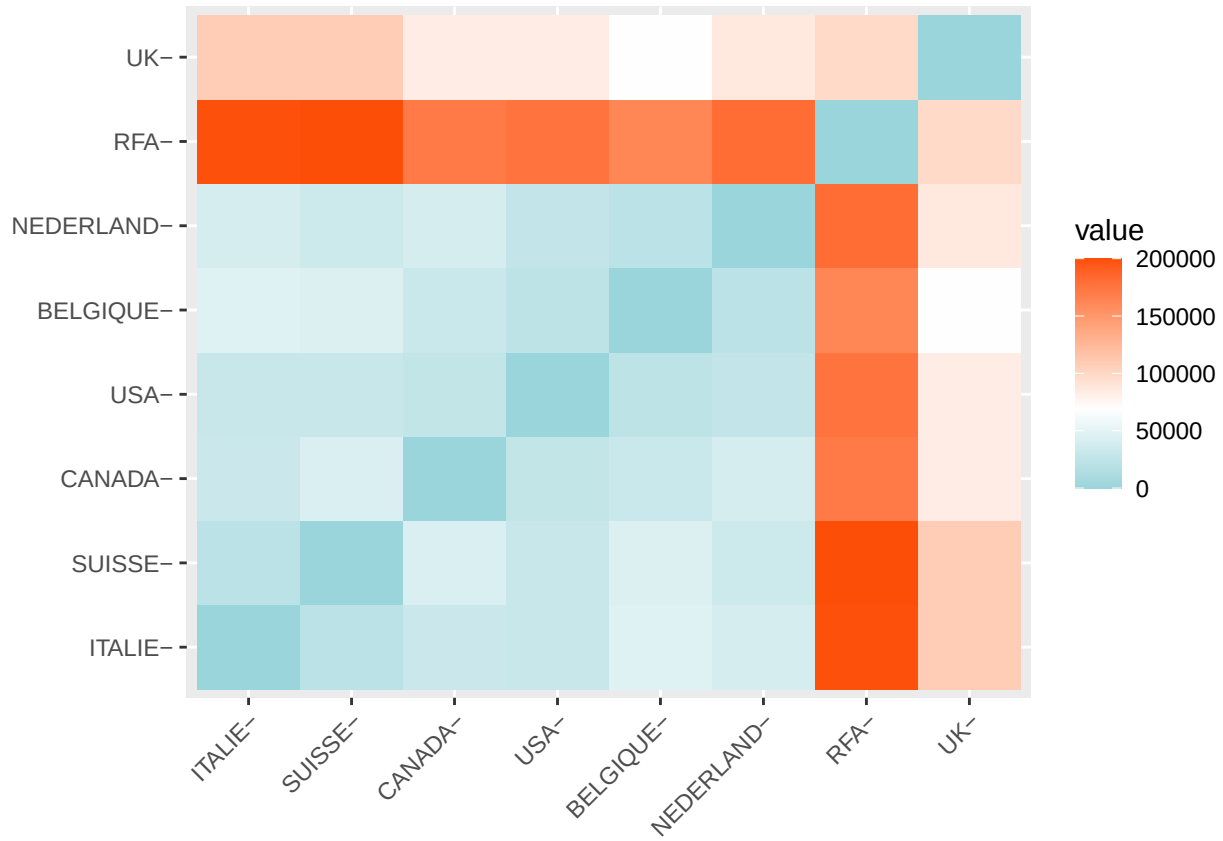



```
CAH$desc.var
```

```
##
## Link between the cluster variable and the quantitative variables
## ================================================================
##                       Eta2      P-value
## ALSACE          0.9989314 2.140176e-06
## AOC_FORT        0.9963153 2.542500e-05
## MUSCAT          0.9933236 8.339130e-05
## PROVENCE        0.9744164 1.216712e-03
## AUTRE_VDQS      0.9624343 2.612601e-03
## AOC_AUTRES      0.9619758 2.676346e-03
## ANJOU           0.9459293 5.381998e-03
## BOJOLAIS        0.9385901 6.924512e-03
## GIRONDE         0.8851958 2.374546e-02
## MOUSSEAU_SIMPLE 0.8741104 2.843738e-02
##
## Description of each cluster by quantitative variables
## =====================================================
```

```
## $`1`
##               v.test Mean in category Overall mean sd in category Overall sd
## RHONE      -2.000474           306.50      680.375       288.4029   494.4729
## VDQS       -2.151191           429.00     1040.625       380.8668   752.2380
## GIRONDE    -2.402317          5790.25    14922.250      4810.8704 10057.3751
## AOC_AUTRES -2.569393          1572.25    10284.750      1480.6969  8971.4211
##               p.value
## RHONE      0.04544910
## VDQS       0.03146112
## GIRONDE    0.01629159
## AOC_AUTRES 0.01018768
##
## $`2`
## NULL
##
## $`3`
##                   v.test Mean in category Overall mean sd in category
## ALSACE          2.617373            17183     3004.625              0
## AUTRE_VDQS      2.318720           191140    52982.750              0
## BOJOLAIS        2.279701            72977    23715.625              0
## PROVENCE        2.156399             2514      746.125              0
## MOUSSEAU_SIMPLE 2.028906            10439     3331.500              0
##                 Overall sd     p.value
## ALSACE           5417.0255 0.008860952
## AUTRE_VDQS      59583.4252 0.020410250
## BOJOLAIS        21608.6970 0.022625409
## PROVENCE          819.8273 0.031052508
## MOUSSEAU_SIMPLE  3503.1199 0.042467894
##
## $`4`
##            v.test Mean in category Overall mean sd in category Overall sd
## AOC_FORT 2.576845             1177      219.875              0   371.4328
## MUSCAT   2.572355            12891     2591.375              0  4003.9673
## ANJOU    2.452823             7582     1752.000              0  2376.8534
##              p.value
## AOC_FORT 0.009970655
## MUSCAT   0.010100929
## ANJOU    0.014174017
```

```
distance <- get_dist(d)
fviz_dist(distance, gradient = list(low = "#00AFBB", mid = "white", high = "#FC4E07"))
```

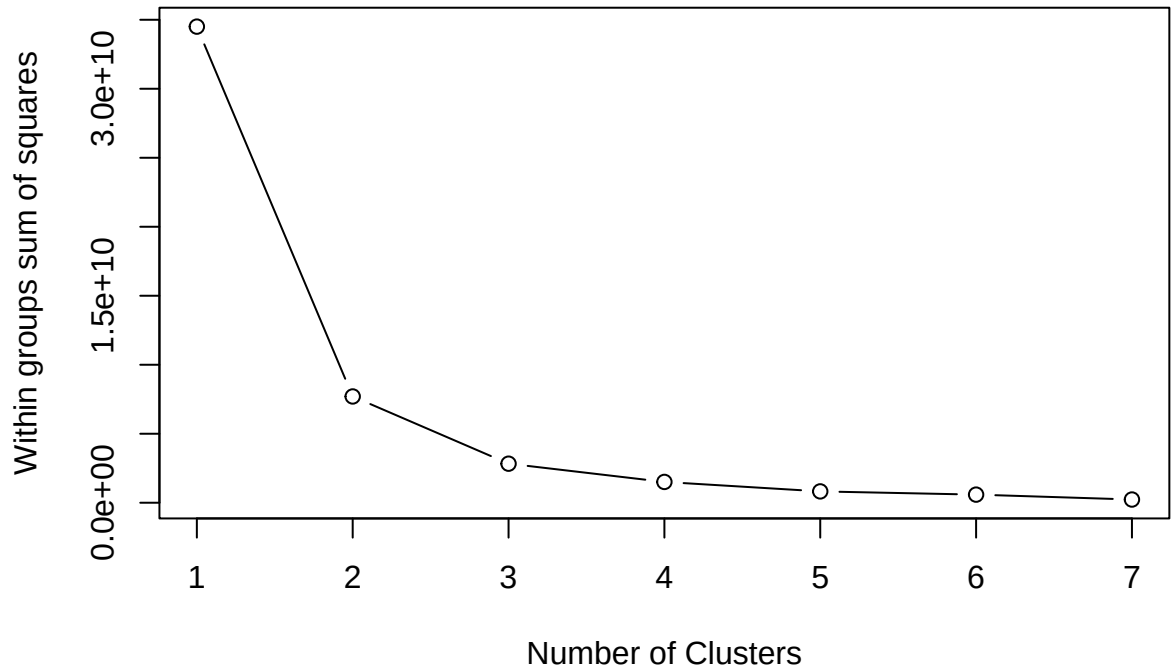



```
library(metan)
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg   ggplot2
## |=========================================================|
## | Multi-Environment Trial Analysis (metan) v1.18.0        |
## | Author: Tiago Olivoto                                   |
## | Type 'citation('metan')' to know how to cite metan      |
## | Type 'vignette('metan_start')' for a short tutorial     |
## | Visit 'https://bit.ly/pkgmetan' for a complete tutorial |
## |=========================================================|
##
## Attachement du package : 'metan'
## L'objet suivant est masqué depuis 'package:factoextra':
##
##     get_dist
```

```
# 2. Choix du nombre de clusters (K) - Exemple avec la méthode du coude
wss <- numeric(7)  # Pour stocker la somme des carrés des distances
for (i in 1:7) {
  wss[i] <- sum(kmeans(d, centers = i)$withinss)
```

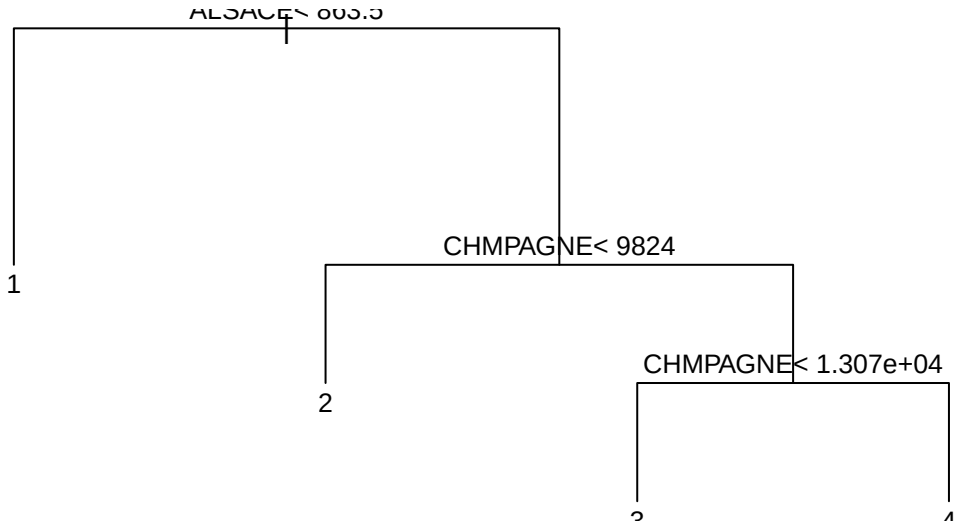
```
}
plot(1:7, wss, type = "b", xlab = "Number of Clusters",
     ylab = "Within groups sum of squares")
```




### K-means

```
k2 <- kmeans(d, centers = 2, nstart = 25)
```

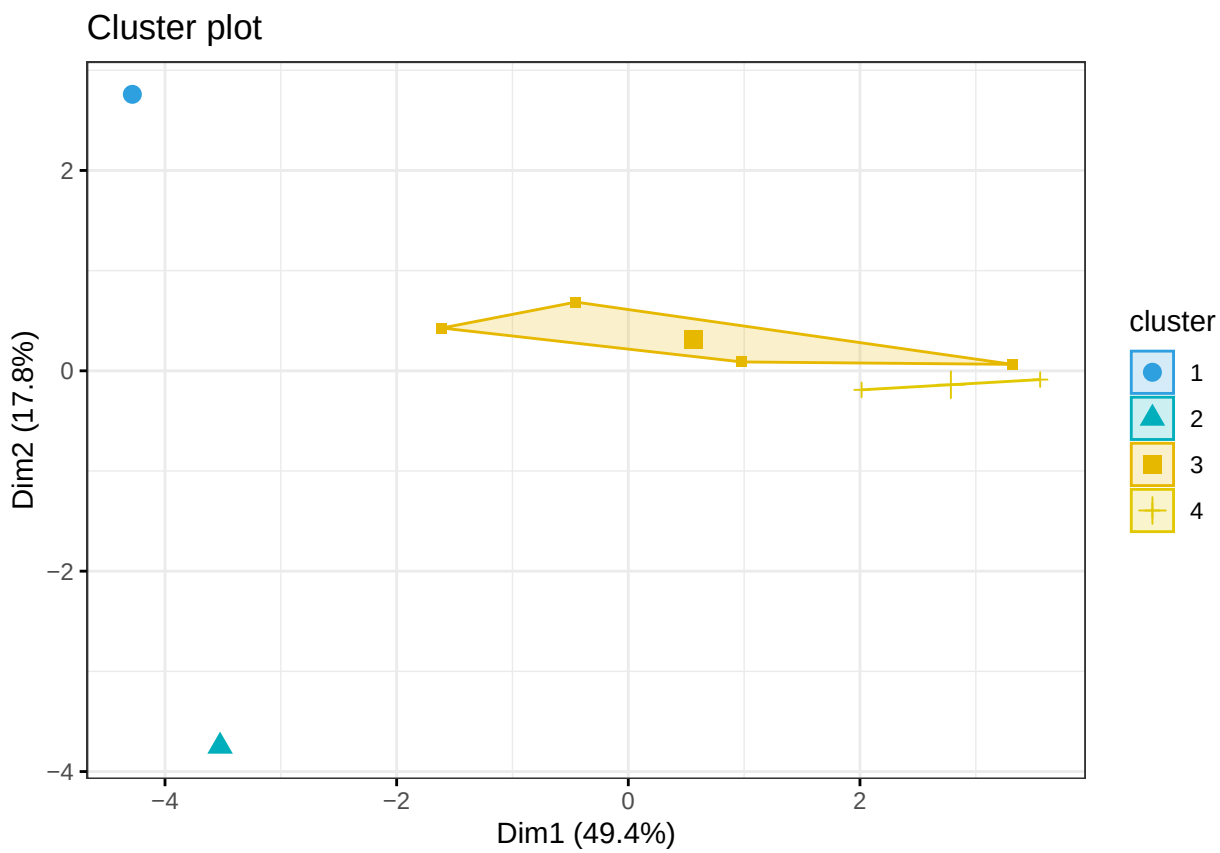
```
library(tree)
dprime=d
nom=rownames(d)
dprime$clust=CAH$data.clust$clust
dprime$clust=as.factor(dprime$clust)
dt=rpart(clust~., data=dprime, minsplit=1)
plot(dt)
text(dt,pretty=0,cex=0.8)
```




```
dt
```

```
## n= 8
##
## node), split, n, loss, yval, (yprob)
##       * denotes terminal node
##
##  1) root 8 4 1 (0.5000000 0.2500000 0.1250000 0.1250000)
##    2) ALSACE< 863.5 4 0 1 (1.0000000 0.0000000 0.0000000 0.0000000) *
##    3) ALSACE>=863.5 4 2 2 (0.0000000 0.5000000 0.2500000 0.2500000)
##      6) CHMPAGNE< 9823.5 2 0 2 (0.0000000 1.0000000 0.0000000 0.0000000) *
##      7) CHMPAGNE>=9823.5 2 1 3 (0.0000000 0.0000000 0.5000000 0.5000000)
##       14) CHMPAGNE< 13067 1 0 3 (0.0000000 0.0000000 1.0000000 0.0000000) *
##       15) CHMPAGNE>=13067 1 0 4 (0.0000000 0.0000000 0.0000000 1.0000000) *
```

```
res.km <- kmeans(d, 4, nstart = 25)
fviz_cluster(res.km, data = d,
             palette = c("#2E9FDF", "#00AFBB", "#E7B800", "#E1C800"),
             geom = "point",
             ellipse.type = "convex",
             ggtheme = theme_bw()
             )
```




```
dprime[,19]=as.integer(dprime[,19])

dprime[,-19]=scale(dprime[,-19])

m1 = lm(clust~., data=dprime)
summary(m1)
```

```
##
## Call:
## lm(formula = clust ~ ., data = dprime)
##
## Residuals:
## ALL 8 residuals are 0: no residual degrees of freedom!
##
## Coefficients: (11 not defined because of singularities)
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)        1.8750        NaN     NaN      NaN
## CHMPAGNE          -0.1718        NaN     NaN      NaN
## MOUSSEAU_AOC      -0.1076        NaN     NaN      NaN
## MOUSSEAU_SIMPLE    0.9578        NaN     NaN      NaN
## ALSACE            -1.1654        NaN     NaN      NaN
## GIRONDE            0.5428        NaN     NaN      NaN
## BOJOLAIS           1.0372        NaN     NaN      NaN
## BORDEAUX          -0.5436        NaN     NaN      NaN
## RHON                   NA         NA      NA       NA
## ANJOU                  NA         NA      NA       NA
```

```
## AOC_AUTRES            NA         NA      NA       NA
## VDQS                  NA         NA      NA       NA
## AUTRE_VDQS            NA         NA      NA       NA
## PROVENCE              NA         NA      NA       NA
## MUSCAT                NA         NA      NA       NA
## RHONE                 NA         NA      NA       NA
## AOC_FORT              NA         NA      NA       NA
## AUTRE_FORT            NA         NA      NA       NA
## TRES_FORT             NA         NA      NA       NA
##
## Residual standard error: NaN on 0 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:    NaN
## F-statistic:   NaN on 7 and 0 DF,  p-value: NA
```

```
library(leaps)
choix = regsubsets(clust~., int= T, nbest = 1, nvmax=7, method="exh", data = dprime)
```

```
## Warning in leaps.setup(x, y, wt = wt, nbest = nbest, nvmax = nvmax, force.in =
## force.in, : 11 linear dependencies found
```

```
res = summary(choix)
```

```
## Warning in log(vr): Production de NaN
```

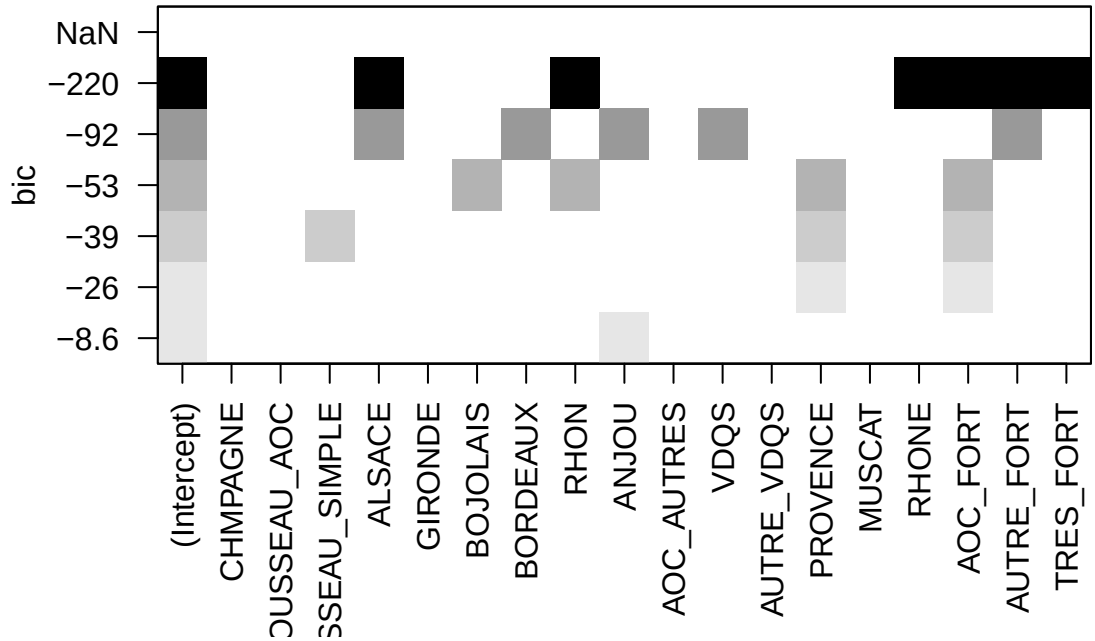
```
print(res)
```

```
## Subset selection object
## Call: regsubsets.formula(clust ~ ., int = T, nbest = 1, nvmax = 7,
##     method = "exh", data = dprime)
## 18 Variables  (and intercept)
##                 Forced in Forced out
## CHMPAGNE            FALSE      FALSE
## MOUSSEAU_AOC        FALSE      FALSE
## MOUSSEAU_SIMPLE     FALSE      FALSE
## ALSACE              FALSE      FALSE
## GIRONDE             FALSE      FALSE
## BOJOLAIS            FALSE      FALSE
## BORDEAUX            FALSE      FALSE
## RHON                FALSE      FALSE
## ANJOU               FALSE      FALSE
## AOC_AUTRES          FALSE      FALSE
## VDQS                FALSE      FALSE
## AUTRE_VDQS          FALSE      FALSE
## PROVENCE            FALSE      FALSE
## MUSCAT              FALSE      FALSE
## RHONE               FALSE      FALSE
## AOC_FORT            FALSE      FALSE
## AUTRE_FORT          FALSE      FALSE
## TRES_FORT           FALSE      FALSE
## 1 subsets of each size up to 7
## Selection Algorithm: exhaustive
##          CHMPAGNE MOUSSEAU_AOC MOUSSEAU_SIMPLE ALSACE GIRONDE BOJOLAIS BORDEAUX
## 1  ( 1 ) " "      " "          " "             " "    " "     " "      " "
## 2  ( 1 ) " "      " "          " "             " "    " "     " "      " "
## 3  ( 1 ) " "      " "          "*"             " "    " "     " "      " "
## 4  ( 1 ) " "      " "          " "             " "    " "     "*"      " "
```

```
## 5  ( 1 ) " "       " "           " "              "*"     " "      " "       "*"
## 6  ( 1 ) " "       " "           " "              "*"     " "      " "       " "
## 7  ( 1 ) "*"       "*"           "*"              "*"     "*"      "*"       " "
##          RHON ANJOU AOC_AUTRES VDQS AUTRE_VDQS PROVENCE MUSCAT RHONE AOC_FORT
## 1  ( 1 ) " "  "*"   " "        " "  " "        " "      " "    " "   " "
## 2  ( 1 ) " "  " "   " "        " "  " "        "*"      " "    " "   "*"
## 3  ( 1 ) " "  " "   " "        " "  " "        "*"      " "    " "   "*"
## 4  ( 1 ) "*"  " "   " "        " "  " "        "*"      " "    " "   "*"
## 5  ( 1 ) " "  "*"   " "        "*"  " "        " "      " "    " "   " "
## 6  ( 1 ) "*"  " "   " "        " "  " "        " "      " "    "*"   "*"
## 7  ( 1 ) " "  " "   " "        " "  " "        " "      " "    " "   " "
##          AUTRE_FORT TRES_FORT
## 1  ( 1 ) " "        " "
## 2  ( 1 ) " "        " "
## 3  ( 1 ) " "        " "
## 4  ( 1 ) " "        " "
## 5  ( 1 ) "*"        " "
## 6  ( 1 ) "*"        "*"
## 7  ( 1 ) "*"        " "
```

```
#choix du meilleur modèle selon le critère BIC
plot(choix, scale="bic")
```

```
## Warning in log(vr): Production de NaN
```




```
m1 = lm(clust~ALSACE+RHON+RHONE+AOC_FORT+AUTRE_FORT+TRES_FORT, data=dprime) # estime le modele linéaire
summary(m1)
```

```
##
## Call:
## lm(formula = clust ~ ALSACE + RHON + RHONE + AOC_FORT + AUTRE_FORT +
##     TRES_FORT, data = dprime)
##
## Residuals:
##   BELGIQUE  NEDERLAND        RFA     ITALIE         UK     SUISSE        USA
##  1.811e-08  2.258e-09  3.987e-09  6.555e-07 -9.446e-09  1.031e-07 -1.140e-07
##     CANADA
## -6.595e-07
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.875e+00  3.333e-07 5625335 1.13e-07 ***
## ALSACE       5.884e-01  5.130e-07 1146900 5.55e-07 ***
## RHON        -2.084e-02  6.338e-07  -32881 1.94e-05 ***
## RHONE       -3.279e-02  1.227e-06  -26727 2.38e-05 ***
## AOC_FORT     8.413e-01  5.142e-07 1636193 3.89e-07 ***
## AUTRE_FORT   1.519e-01  9.663e-07  157219 4.05e-06 ***
## TRES_FORT    2.028e-01  5.285e-07  383665 1.66e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.428e-07 on 1 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 1.664e+12 on 6 and 1 DF,  p-value: 5.934e-07
```

```
library(performance)
check_collinearity(m1)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##       Term  VIF     VIF 95% CI Increased SE Tolerance Tolerance 95% CI
##     ALSACE 2.07 [ 2.07,  2.07]         1.44      0.48     [0.48, 0.48]
##       RHON 3.16 [ 3.16,  3.16]         1.78      0.32     [0.32, 0.32]
##   AOC_FORT 2.08 [ 2.08,  2.08]         1.44      0.48     [0.48, 0.48]
##  TRES_FORT 2.20 [ 2.20,  2.20]         1.48      0.45     [0.45, 0.45]
##
## Moderate Correlation
##
##        Term  VIF     VIF 95% CI Increased SE Tolerance Tolerance 95% CI
##  AUTRE_FORT 7.35 [ 7.35,  7.35]         2.71      0.14     [0.14, 0.14]
##
## High Correlation
##
##   Term   VIF     VIF 95% CI Increased SE Tolerance Tolerance 95% CI
##  RHONE 11.85 [11.85, 11.85]         3.44      0.08     [0.08, 0.08]
```

```
m2 <- update(m1, .~.-RHONE)
check_collinearity(m2)
```

```
## # Check for Multicollinearity
##
```

```
## Low Correlation
##
##        Term  VIF   VIF 95% CI Increased SE Tolerance Tolerance 95% CI
##      ALSACE 1.10 [1.04, 1.24]         1.05      0.91     [0.80, 0.96]
##        RHON 1.62 [1.47, 1.81]         1.27      0.62     [0.55, 0.68]
##    AOC_FORT 1.77 [1.60, 1.98]         1.33      0.57     [0.50, 0.63]
##  AUTRE_FORT 1.64 [1.49, 1.84]         1.28      0.61     [0.54, 0.67]
##   TRES_FORT 1.75 [1.59, 1.96]         1.32      0.57     [0.51, 0.63]
```

```
summary(m2)
```

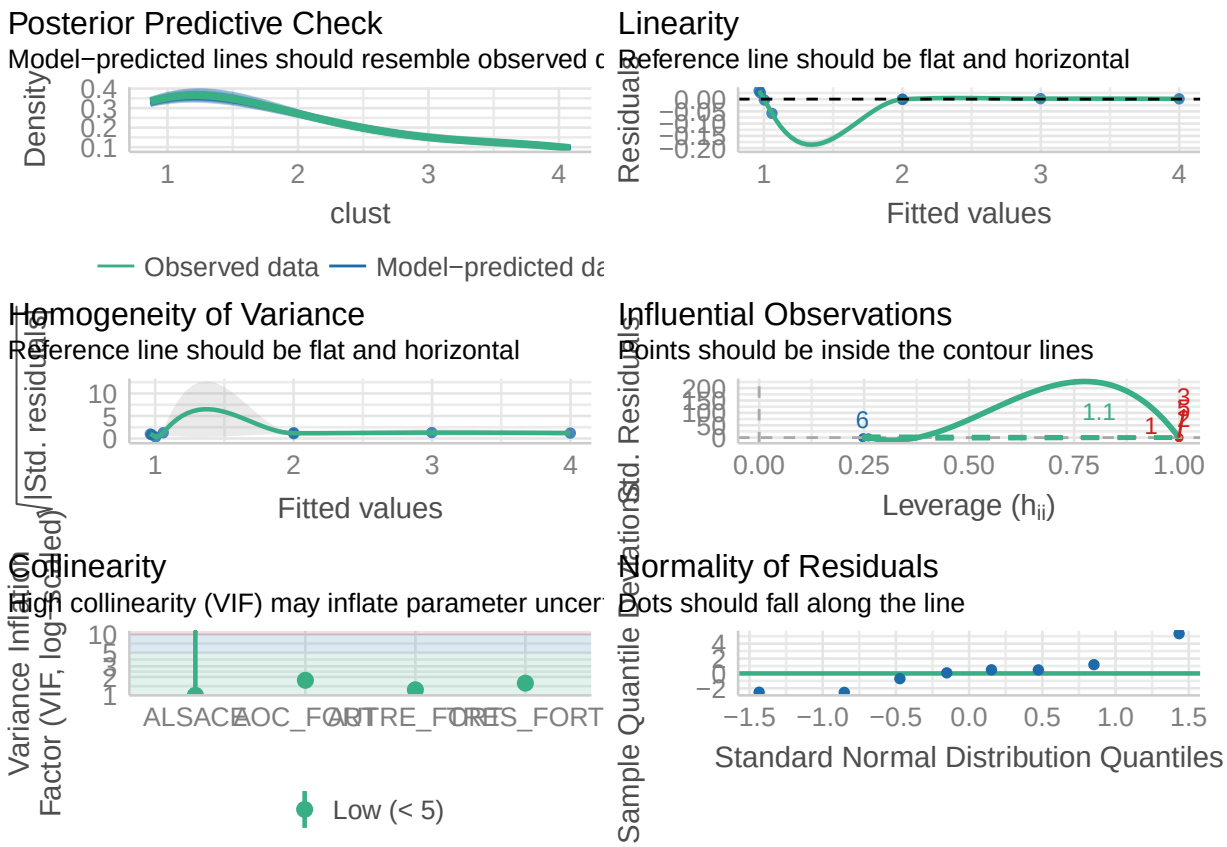
```
##
## Call:
## lm(formula = clust ~ ALSACE + RHON + AOC_FORT + AUTRE_FORT +
##     TRES_FORT, data = dprime)
##
## Residuals:
##   BELGIQUE  NEDERLAND        RFA     ITALIE         UK     SUISSE        USA
##  2.801e-06 -4.035e-05  2.863e-04  8.201e-03  3.330e-05  4.191e-04 -2.070e-02
##     CANADA
##  1.179e-02
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.875000   0.006299 297.654 1.13e-05 ***
## ALSACE       0.578984   0.007055  82.065 0.000148 ***
## RHON        -0.032689   0.008563  -3.818 0.062277 .
## AOC_FORT     0.846666   0.008951  94.585 0.000112 ***
## AUTRE_FORT   0.129165   0.008629  14.969 0.004433 **
## TRES_FORT    0.196393   0.008916  22.026 0.002055 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01782 on 2 degrees of freedom
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9997
## F-statistic:  5591 on 5 and 2 DF,  p-value: 0.0001788
```

```
m2 <- update(m2, .~.-RHON)
summary(m2)
```

```
##
## Call:
## lm(formula = clust ~ ALSACE + AOC_FORT + AUTRE_FORT + TRES_FORT,
##     data = dprime)
##
## Residuals:
##  BELGIQUE NEDERLAND       RFA    ITALIE        UK    SUISSE       USA    CANADA
## -0.001220 -0.000525  0.001793  0.034569  0.001140 -0.058153 -0.003419  0.025815
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.87500    0.01481 126.638 1.09e-06 ***
## ALSACE       0.57109    0.01585  36.023 4.70e-05 ***
## AOC_FORT     0.84682    0.02104  40.250 3.37e-05 ***
## AUTRE_FORT   0.11273    0.01758   6.413  0.00768 **
```

```
## TRES_FORT    0.18589    0.01994   9.325  0.00261 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04188 on 3 degrees of freedom
## Multiple R-squared:  0.9994, Adjusted R-squared:  0.9986
## F-statistic:  1264 on 4 and 3 DF,  p-value: 3.607e-05
```

```
library(see)
check_model(m2)
```




```
check_normality(m2)
```

```
## OK: residuals appear as normally distributed (p = 0.250).
```

```
check_heteroscedasticity(m2)
```

```
## OK: Error variance appears to be homoscedastic (p = 0.108).
```